

LOW DROP POWER SCHOTTKY RECTIFIER

MAIN PRODUCTS CHARACTERISTICS

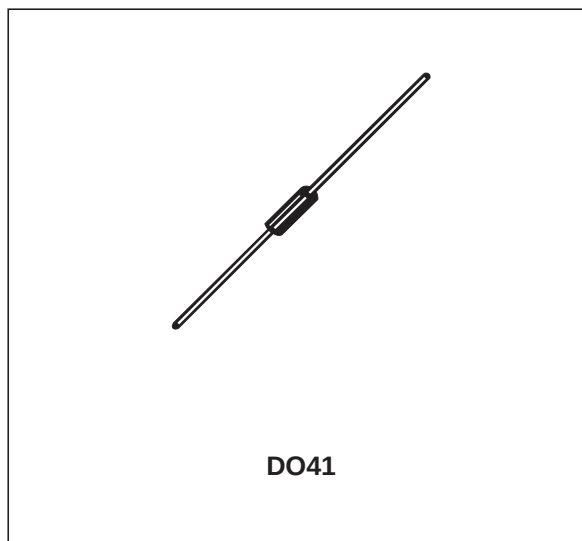
$I_F(AV)$	1 A
V_{RRM}	40 V
T_j	150°C
$V_F(max)$	0.45 V

FEATURES AND BENEFITS

- VERY SMALL CONDUCTION LOSSES
- NEGLIGIBLE SWITCHING LOSSES
- EXTREMELY FAST SWITCHING
- LOW FORWARD VOLTAGE DROP
- AVALANCHE CAPABILITY SPECIFIED

DESCRIPTION

Axial Power Schottky rectifier suited for Switch Mode Power Supplies and high frequency DC to DC converters. Packaged in DO41 these devices are intended for use in low voltage, high frequency inverters, free wheeling, polarity protection and small battery chargers.



ABSOLUTE RATINGS (limiting values)

Symbol	Parameter		Value			Unit
			1N5817	1N5818	1N5819	
V_{RRM}	Repetitive peak reverse voltage		20	30	40	V
$I_{F(RMS)}$	RMS forward current		10			A
$I_{F(AV)}$	Average forward current	$T_L = 125^\circ\text{C}$ $\delta = 0.5$	1			A
I_{FSM}	Surge non repetitive forward current	$t_p = 10\text{ ms}$ Sinusoidal	25			A
P_{ARM}	Repetitive peak avalanche power	$t_p = 1\mu\text{s}$ $T_j = 25^\circ\text{C}$	1200	1200	900	W
T_{stg}	Storage temperature range		- 65 to + 150			°C
T_j	Maximum operating junction temperature *		150			°C
dV/dt	Critical rate of rise of reverse voltage		10000			V/ μs

* : $\frac{dP_{tot}}{dT_j} < \frac{1}{R_{th}(j - a)}$ thermal runaway condition for a diode on its own heatsink

1N581x

THERMAL RESISTANCES

Symbol	Parameter		Value	Unit
$R_{th(j-a)}$	Junction to ambient	Lead length = 10 mm	100	°C/W
$R_{th(j-l)}$	Junction to lead	Lead length = 10 mm	45	°C/W

STATIC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	Tests Conditions		1N5817	1N5818	1N5819	Unit
I_R^*	Reverse leakage current	$T_j = 25^\circ\text{C}$	$V_R = V_{RRM}$	0.5	0.5	0.5	mA
		$T_j = 100^\circ\text{C}$		10	10	10	mA
V_F^*	Forward voltage drop	$T_j = 25^\circ\text{C}$	$I_F = 1\text{ A}$	0.45	0.50	0.55	V
		$T_j = 25^\circ\text{C}$	$I_F = 3\text{ A}$	0.75	0.80	0.85	V

Pulse test : * $t_p = 380\text{ }\mu\text{s}$, $\delta < 2\%$

To evaluate the conduction losses use the following equations :

$$P = 0.3 \times I_{F(AV)} + 0.090 I_{F(RMS)}^2 \text{ for 1N5817 / 1N5818}$$

$$P = 0.3 \times I_{F(AV)} + 0.150 I_{F(RMS)}^2 \text{ for 1N5819}$$

Fig. 1: Average forward power dissipation versus average forward current (1N5817/1N5818).

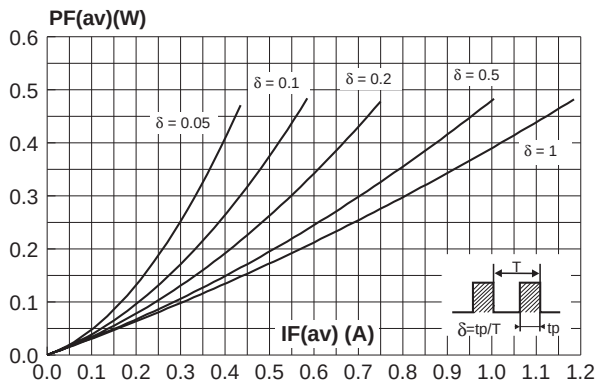


Fig. 2-1: Average forward current versus ambient temperature ($\delta=0.5$) (1N5817/1N5818).

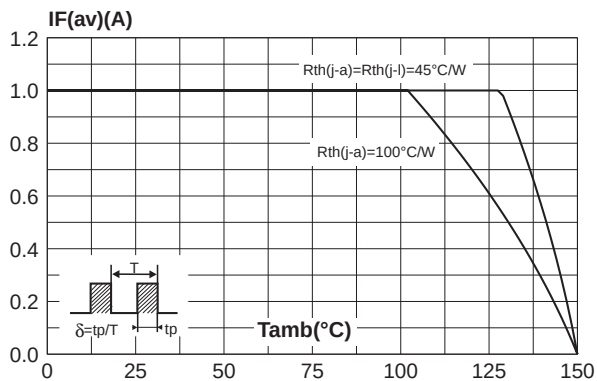


Fig. 2: Average forward power dissipation versus average forward current (1N5819).

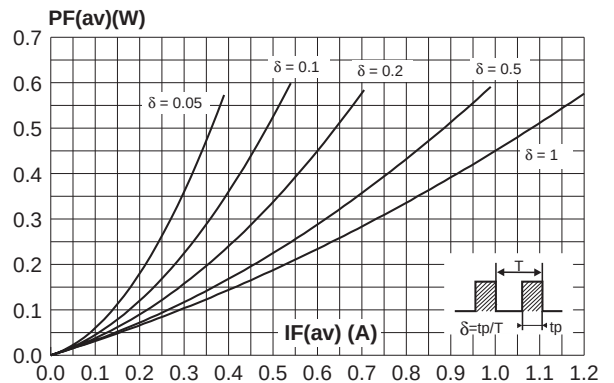


Fig. 2-2: Average forward current versus ambient temperature ($\delta=0.5$) (1N5819).

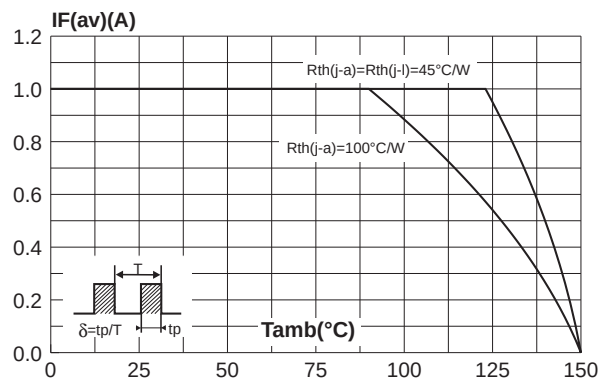


Fig. 3: Normalized avalanche power derating versus pulse duration.

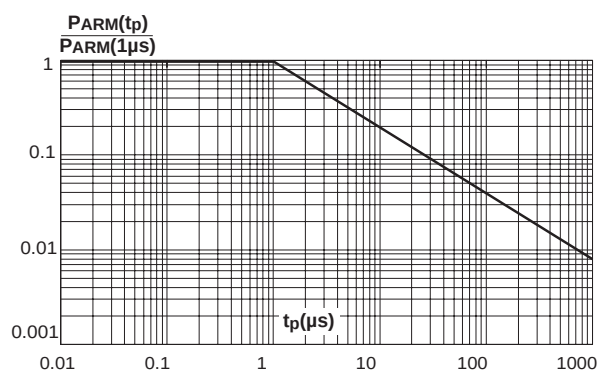


Fig. 4: Normalized avalanche power derating versus junction temperature.

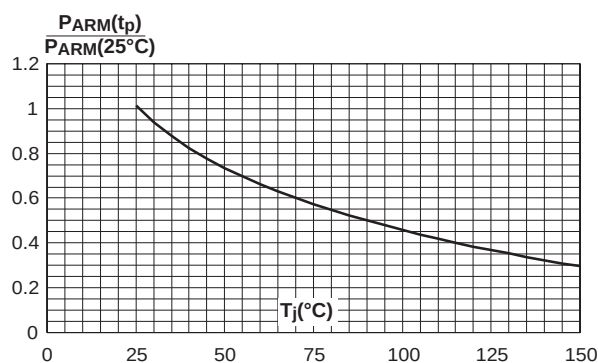


Fig. 5-1: Non repetitive surge peak forward current versus overload duration (maximum values) (1N5817/1N5818).

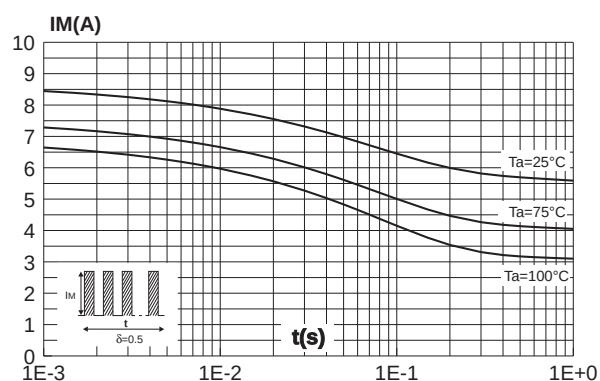


Fig. 5-2: Non repetitive surge peak forward current versus overload duration (maximum values) (1N5819).

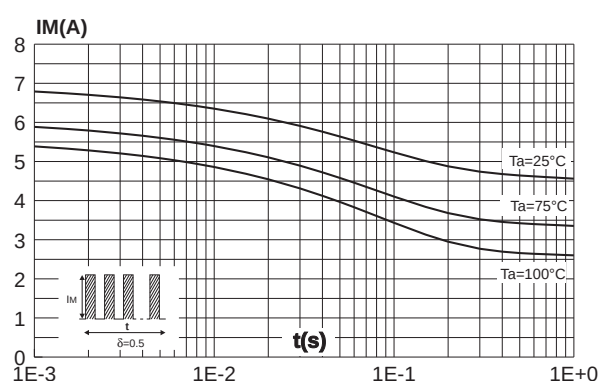


Fig. 6: Relative variation of thermal impedance junction to ambient versus pulse duration (epoxy printed circuit board, e(Cu)=35mm, recommended pad layout).

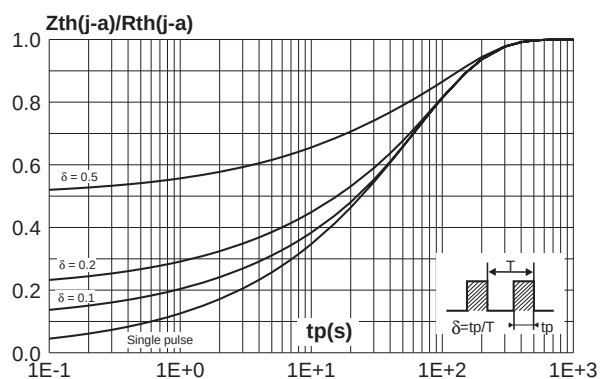


Fig. 7: Junction capacitance versus reverse voltage applied (typical values).

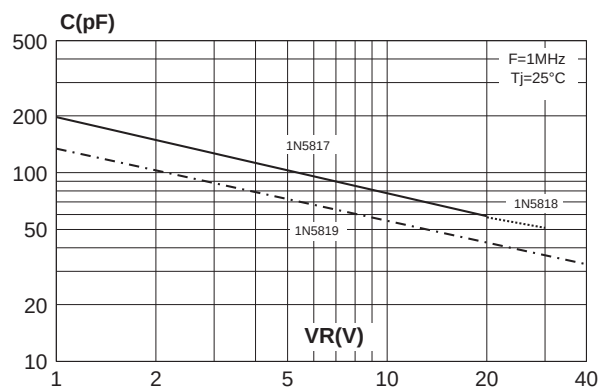


Fig. 8-1: Reverse leakage current versus reverse voltage applied (typical values) (1N5817/1N5818).

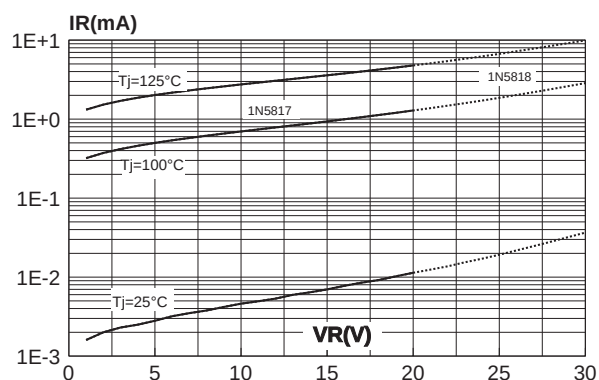


Fig. 8-2: Reverse leakage current versus reverse voltage applied (typical values) (1N5819).

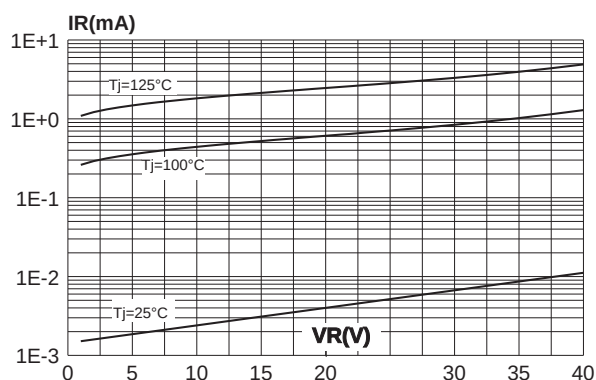


Fig. 9-1: Forward voltage drop versus forward current (typical values) (1N5817/1N5818).

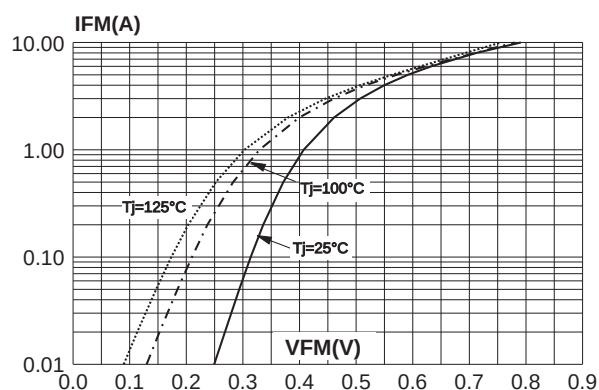


Fig. 9-2: Forward voltage drop versus forward current (typical values) (1N5819).

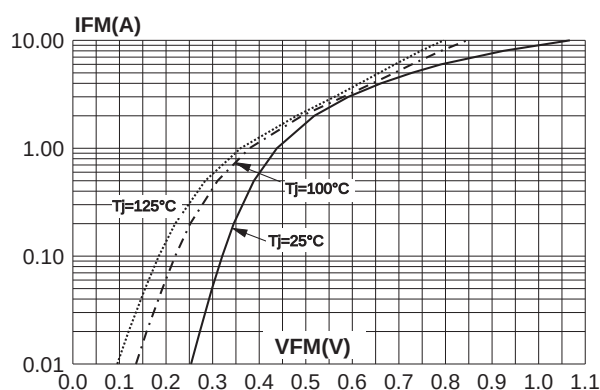
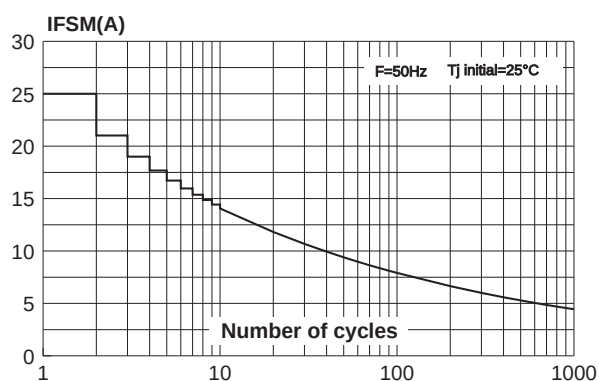


Fig. 10: Non repetitive surge peak forward current versus number of cycles.



PACKAGE MECHANICAL DATA

DO41 plastic

REF.	DIMENSIONS			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
A	4.1	5.2	0.16	0.205
B	2	2.7	0.08	0.107
C	25.4		1	
D	0.71	0.86	0.028	0.034

Ordering type	Marking	Package	Weight	Base qty	Delivery mode
1N581x	Part number cathode ring	DO41	0.34g	2000	Ammopack
1N581xRL	Part number cathode ring	DO41	0.34g	5000	Tape & reel

- Epoxy meets UL94,V0

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